University of Science and Technology Houari Boumediene
Faculty of Computer Science
Bachelor 2 (L2) - ACAD (A, B and C) Academic year: 2026/2027
Module: Probability and Statistics

Tutorial Sheet No. 1
Combinatorics

Exercise 1: How many four-letter company acronyms can be formed from the letters of the Latin alphabet?

Exercise 2: Using the letters of the word SCIENCES, how many eight-letter words, meaningful or not, can be formed?

Exercise 3: How many three-digit numbers with all digits different can be formed from the digits $\{1, 2, 3, \ldots, 9\}$?
Similarly, how many three-digit numbers with all digits different can be formed from the digits $\{0, 1, 2, 3, \ldots, 9\}$?

Exercise 4: Calculate the number of subsets of a set with n elements.

Exercise 5: Calculate the number of diagonals of a convex polygon with n sides.

Exercise 6: In a bank, each customer has an account whose code consists of 3 letters and 5 digits, not necessarily distinct, of the type R G A 4 9 2 8 1.

1) Assume that the 3 letters are distinct.
   How many accounts can be opened whose code:
   a) Starts with A B?
   b) Starts with A?
   c) Contains an A?
   d) Contains an A and a B?
   e) Starts with A and ends with 1 2 3?
2) Assume that the 3 letters are no longer necessarily distinct.
   How many accounts can be opened whose code:
   a) Starts with A?
   b) Contains at least two A's?
3) Assume that the 3 letters are not necessarily distinct and that the digits 0, 1, 2, 3, 4 cannot be used.
   How many accounts can be opened whose code:
   a) Starts with A?
   b) Ends with 999?
   c) Starts with A and ends with 89?

Exercise 7: We have n_1 white cubes and n_2 red cubes. They are placed at random in boxes numbered from 1 to $n_1 + n_2$.

1) What is the number of distinct possible arrangements?
2) What is the number of arrangements when k specific white cubes are in fixed positions?

Exercise 8: In a class of 32 students, there are 19 boys and 13 girls. Two class representatives must be elected:

1) What is the number of possible choices?
2) What is the number of choices if one boy and one girl are required?
3) What is the number of choices if 2 boys are required?

Exercise 9: A bag contains 5 green tokens (numbered from 1 to 5) and 4 red tokens (numbered from 1 to 4).

1) 3 tokens are drawn successively and at random from the bag, without replacement. Calculate the number of possibilities of:
   i) Drawing only 3 green tokens.
   ii) Drawing no green token.
   iii) Drawing at most 2 green tokens.
   iv) Drawing exactly 1 green token.
2) 3 tokens are drawn simultaneously and at random from the bag. Redo the previous questions.

Exercise 10: Two dice numbered from 1 to 6 are rolled simultaneously. Determine the sample space Ω (the set of all possible outcomes) in the following cases:

a) The two dice are distinguishable (for example, one red and one blue).
b) The two dice are identical.
c) The two dice are identical and we are only interested in the parity of the result.